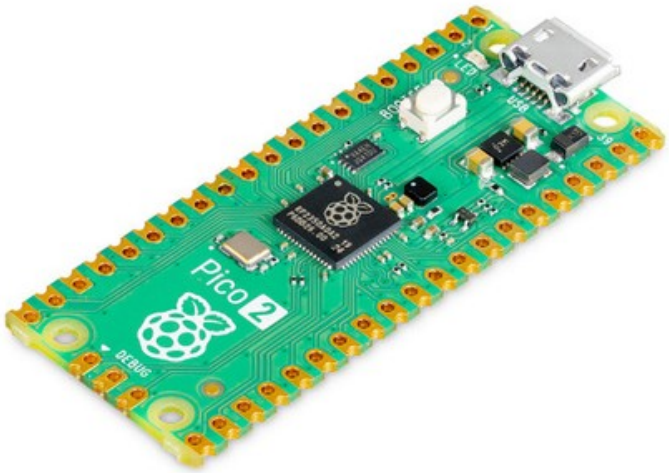
**********Draft**********

High Speed Data Acquisition Over HDMI hsdaoh

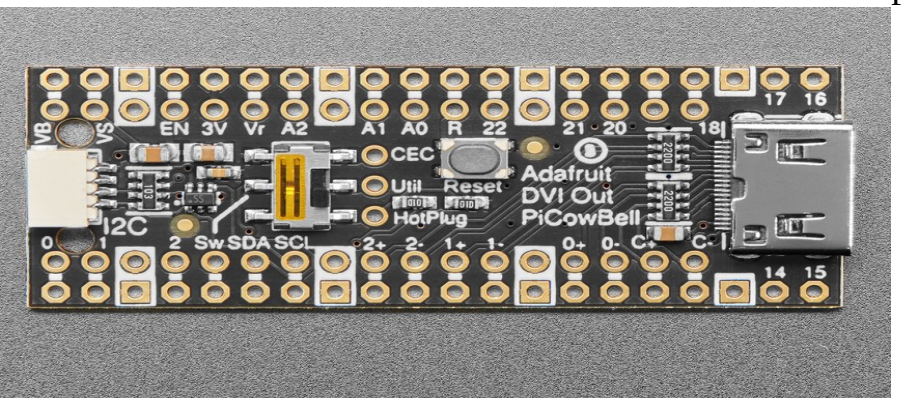
10/30/26

**********Draft**********

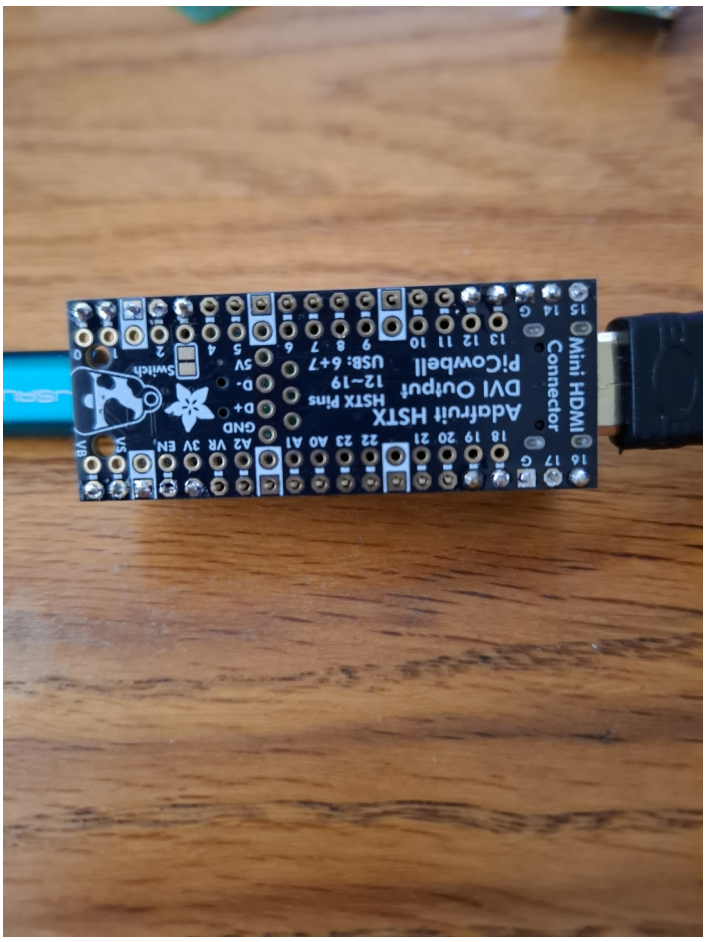
pico2



PicowBell-dvi



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Test pico2 with PicowBell-dvi

https://github.com/develone/pico-examples/blob/test-dev/doc/hstx/mtn-hstx-wkg-091426.jpg

https://github.com/develone/pico-examples/blob/test-dev/doc/hstx/lena_640x480_rgb332.h.jpg

The information below was found at “https://www.electronics-lab.com/news/the-ms2130-mini-featuring-the-raspberry-pi-pico-2s-hstx-interface-is-an-hdmi-to-usb-3-0-4k-video-capture-card-that-supports-1080p-60fps-for-live-streaming-and-recording/”
xx

4K Video Capture Card, Cam Link Card Game Capture Card Audio Capture Adapter HDMI to USB 2.0 Record Capture Device for Streaming, Live Broadcasting, Video Conference, Teaching, Gaming(Black)

4.3*4.3 out of 5 stars*
(3.2K)
100+ bought in past month
Price, product page$9.98



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The MS2130 Mini, featuring the Raspberry Pi Pico 2's HSTX interface, is an HDMI to USB 3.0 4K video capture card that supports 1080p 60fps for live streaming and recording.

The MS2130 Mini 4K Video Capture Card is a compact and high-quality device designed for capturing HDMI video and audio signals for use on computers or online streaming platforms. It supports input resolutions up to 4K@30Hz and outputs at 1080P@60fps with YUY2 full HD color depth, ensuring sharp, detailed visuals.

The above site also had a link to "https://github.com/steve-m/hsdaoh-rp2350" for the Raspberry Pi pico2.

To build hsdaoh-rp2350: After creating a fork of the repo and the test-dev branch.

```
devel@pi5-80:~/hsdaoh-rp2350/build $ git clone -b test-dev
https://github.com/develone/hsdaoh-rp2350.git

devel@pi5-80:~/hsdaoh-rp2350/build $ mkdir hsdaoh-rp2350/build

devel@pi5-80:~/hsdaoh-rp2350/build $ cd hsdaoh-rp2350/build

devel@pi5-80:~/hsdaoh-rp2350/build $ cmake -DPICO_PLATFORM=rp2350 -
DPICO_BOARD=pico2 ../
```

```
devel@pi5-80:~/hsdaoh-rp2350/build $ make -j 8
```

devel@pi5-80:~/hsdaoh-rp2350/build $ ls apps/counter/
CMakeFiles counter.bin counter.elf counter.hex counter.uf2
cmake_install.cmake counter.dis counter.elf.map counter.pio.h Makefile

needed additional packages

devel@pi5-80:~ $ sudo apt install flac libflac-dev libuvc-dev v4l-utils guvcview
devel@pi5-80:~ $ git clone https://github.com/develone/hsdaoh.git
devel@pi5-80:~/hsdaoh $ mkdir hsdaoh/build
devel@pi5-80:~ $ cd hsdaoh/build/
devel@pi5-80:~/hsdaoh/build $ cmake ../ -DINSTALL_UDEV_RULES=ON
devel@pi5-80:~/hsdaoh/build $ make -j 4
[20%] Building C object src/CMakeFiles/hsdaoh.dir/format_convert.c.o
[20%] Building C object src/CMakeFiles/hsdaoh_static.dir/libhsdaoh.c.o
[20%] Building C object src/CMakeFiles/hsdaoh.dir/iqconverter_float.c.o
[20%] Building C object src/CMakeFiles/hsdaoh.dir/libhsdaoh.c.o
[25%] Building C object src/CMakeFiles/hsdaoh.dir/crcspeed.c.o
[30%] Building C object src/CMakeFiles/hsdaoh_static.dir/format_convert.c.o
[35%] Building C object src/CMakeFiles/hsdaoh.dir/crc16speed.c.o
[40%] Building C object src/CMakeFiles/hsdaoh.dir/crc_simd.c.o
[45%] Building C object src/CMakeFiles/hsdaoh_static.dir/iqconverter_float.c.o
[50%] Building C object src/CMakeFiles/hsdaoh_static.dir/crcspeed.c.o
[55%] Building C object src/CMakeFiles/hsdaoh_static.dir/crc16speed.c.o
[60%] Building C object src/CMakeFiles/hsdaoh_static.dir/crc_simd.c.o
[65%] Linking C shared library libhsdaoh.so
[65%] Built target hsdaoh
[70%] Building C object src/CMakeFiles/hsdaoh_file.dir/hsdaoh_file.c.o
[75%] Building C object src/CMakeFiles/hsdaoh_tcp.dir/hsdaoh_tcp.c.o
[80%] Building C object src/CMakeFiles/hsdaoh_test.dir/hsdaoh_test.c.o
[85%] Linking C static library libhsdaoh.a
[90%] Linking C executable hsdaoh_test
[90%] Built target hsdaoh_test
[95%] Linking C executable hsdaoh_tcp
[100%] Linking C executable hsdaoh_file
[100%] Built target hsdaoh_tcp
[100%] Built target hsdaoh_file
[100%] Built target hsdaoh_static

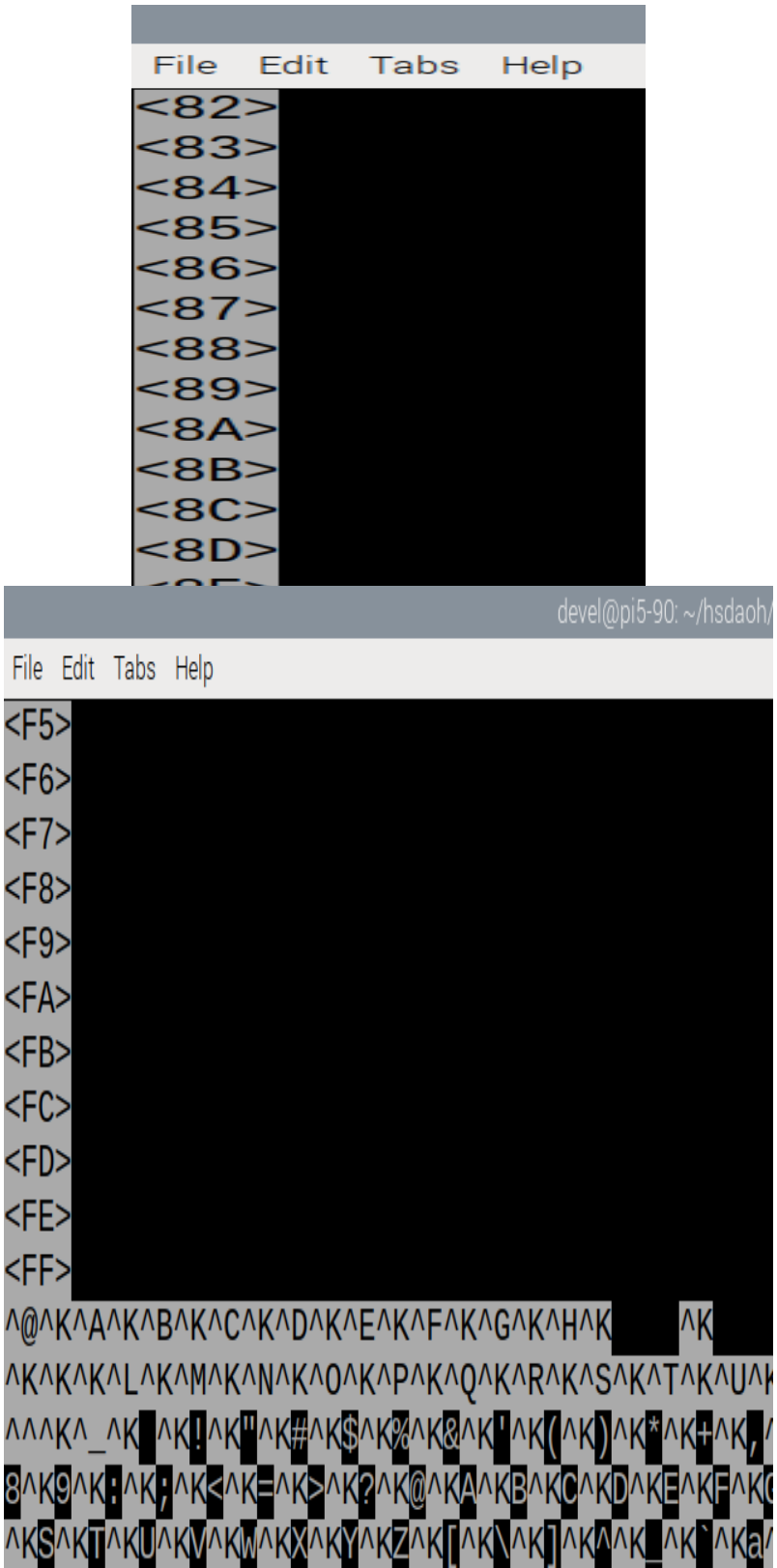
devel@pi5-80:~/hsdaoh/build $ sudo make install
[35%] Built target hsdaoh
[70%] Built target hsdaoh_static
[80%] Built target hsdaoh_file
[90%] Built target hsdaoh_tcp
[100%] Built target hsdaoh_test
Install the project...
-- Install configuration: "Release"
-- Installing: /etc/udev/rules.d/71-hsdaoh.rules
-- Installing: /usr/local/include/hsdaoh.h
-- Installing: /usr/local/include/hsdaoh_export.h

```
-- Installing: /usr/local/lib/pkgconfig/libhsdaoh.pc
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohTargets.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohTargets-release.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohConfig.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohConfigVersion.cmake
-- Installing: /usr/local/lib/libhsdaoh.so.0.1.0
-- Installing: /usr/local/lib/libhsdaoh.so.0
-- Installing: /usr/local/lib/libhsdaoh.so
-- Installing: /usr/local/lib/libhsdaoh.a
-- Installing: /usr/local/bin/hsdaoh_file
-- Set runtime path of "/usr/local/bin/hsdaoh_file" to ""
-- Installing: /usr/local/bin/hsdaoh_tcp
-- Set runtime path of "/usr/local/bin/hsdaoh_tcp" to ""
-- Installing: /usr/local/bin/hsdaoh_test
-- Set runtime path of "/usr/local/bin/hsdaoh_test" to ""
devel@pi5-80:~/hsdaoh/build $ sudo ldconfig
devel@pi5-80:~/hsdaoh/build $ cp ~/hsdaoh-rp2350/build/apps/counter/counter.uf2
/media/devel/RP2350/
devel@pi5-80:~/hsdaoh/build $ hsdaoh_test
unsupported descriptor subtype VS_COLORFORMAT
unsupported descriptor subtype VS_COLORFORMAT
Reporting PPM error measurement every 10 seconds...
Press ^C after a few minutes.
attempt to claim already-claimed interface 1
Synchronized to HDMI input stream
real sample rate: 64003365 current PPM: 1133446 cumulative PPM: 1133446
real sample rate: 63947348 current PPM: 1131578 cumulative PPM: 1132502

devel@pi5-90:~/hsdaoh/build $ hsdaoh_test
unsupported descriptor subtype VS_COLORFORMAT
unsupported descriptor subtype VS_COLORFORMAT
Reporting PPM error measurement every 10 seconds...
Press ^C after a few minutes.
attempt to claim already-claimed interface 1
Synchronized to HDMI input stream
real sample rate: 64023817 current PPM: 1134127 cumulative PPM: 1134127
real sample rate: 63957098 current PPM: 1131903 cumulative PPM: 1132980
real sample rate: 64048601 current PPM: 1134953 cumulative PPM: 1133652
real sample rate: 63952631 current PPM: 1131754 cumulative PPM: 1133169
:

devel@pi5-90:~/hsdaoh $ hsdaoh_test
unsupported descriptor subtype VS_COLORFORMAT
unsupported descriptor subtype VS_COLORFORMAT
Reporting PPM error measurement every 10 seconds...
Press ^C after a few minutes.
attempt to claim already-claimed interface 1
Synchronized to HDMI input stream
real sample rate: 64004096 current PPM: 1133470 cumulative PPM: 1133470
real sample rate: 63939347 current PPM: 1131312 cumulative PPM: 1132380
real sample rate: 64064408 current PPM: 1135480 cumulative PPM: 1133419
^CSignal caught, exiting!
```

devel@pi5-90:~/hsdaoh $ devel@pi5-90:~/hsdaoh $ hsdaoh_file > counter.txt



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